

Performance & Unit Testing Report

1. Model Performance Results

Decision Tree yields 96.55% test accuracy.

- F1-Score: 96.55%
- Precision: 100% (0 False Positives)
- Recall: 94.12% (1 False Negative)

2. Automated Unit Tests Summary

Running test_vercel_app.py evaluates 10 unit test cases:

- test_home_page (200 OK)
- test_predictor_page (200 OK)
- test_docs_page (200 OK)
- test_portal_page (200 OK)
- test_predict_low_risk (Success - Low Risk)
- test_predict_high_risk (Success - High Risk)
- test_validation_negative_input (400 Bad Request)
- test_validation_range_cloud_cover (400 Bad Request)
- test_validation_logical_check (400 Bad Request)
- test_api_docs_list (200 OK)

All 10 tests passed successfully in 0.086 seconds.